Duty-Cycled Shell Sampling: Turning Redundant Evaluations of Grey Wolf Optimization into Productive Exploration

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***Abstract*—Grey wolf optimization (GWO) spends most of its evaluation budget the way an attraction-driven swarm inevitably does: once the pack contracts onto a promising basin, every generation evaluates a tight cluster of nearly identical positions. On CEC2017 at a fixed budget of real objective evaluations we audit this behaviour and find that standard GWO spends about 86% of its budget inside stagnation stretches of at least 1% of the budget, recovers from those stretches more slowly than strong particle-swarm variants, and converts only about 1% of its novel evaluations into objective improvements - roughly ten times less than dynamic multi-swarm PSO. This paper turns that observation into a deliberately small algorithmic change. Duty-cycled shell sampling (DCS-GWO) keeps the GWO update equations and the evaluation budget of every generation, but re-assigns a fixed share of that generation's evaluations from leader-following moves to *shell sampling*: the least productive wolves sample randomly inside a search-space shell centred on the incumbent solution, whose radius is learned online from the median length of recently *successful* displacements. No additional true evaluations, surrogate models, restarts or hand-tuned schedule are introduced. On the CEC2017 benchmark at $D = 50$ with 100,000 true evaluations, 30 independent runs and 29 functions, DCS-GWO reduces the median error ratio against standard GWO to 0.79, wins 21 of 29 functions with eleven Holm-significant improvements and no significant losses, and raises the fraction of novel evaluations that connect to real improvements by 64strongest heterogeneous PSO variants (median ratios 1.9-2.0), which we report explicitly. A pre-registered dual-axis protocol additionally reports a negative result: stagnation triggered re-deployment variants (far novelty, random, elite-memory and trust-region expansion targets) do not improve the operational recovery probability of GWO under the same budget, whereas the duty-cycled shell sampling improves per-evaluation productivity. We therefore report the exploration-efficiency axis as supported and the stagnation-recovery axis as not supported at the mechanism level.**

***Index Terms*—Grey wolf optimizer, evaluation efficiency, exploration–exploitation, swarm intelligence, CEC2017, benchmarking.**

I. Introduction

EVOLUTIONARY and swarm-intelligence algorithms are routinely applied to black-box optimization problems in which a single objective evaluation is expensive, while arithmetic on candidate solutions is cheap. Under this asymmetry the true evaluation budget - not CPU time - is the resource that decides solution quality. Most population-based algorithms, including the grey wolf optimizer (GWO) [**?**], nevertheless carry an implicit design assumption inherited from uncostly-evaluation settings: *every candidate produced by the population is evaluated, once per generation.*

GWO drives its population through attraction toward the three best wolves (α, β, δ) with a linearly decaying exploration parameter a. The mechanism is elegant and has produced a very large derivative literature [**?**]. Its evaluation behaviour, however, has a structural weakness that has received less attention than its search equations: as the pack contracts, the candidate positions generated by the N attraction moves concentrate in a vanishing neighbourhood of the incumbent, and the N evaluations spent per generation degenerate into redundant confirmations of an already-known point. Candidate-generation arguments such as "more diverse leaders" or "additional mutation" do not address this: the waste is in *which positions receive the scarce true evaluations*, not only in how positions are generated.

This paper starts from a measurement of that waste. Using a common evaluation-interface probe for every algorithm (Section III-B), we quantify, at identical real-evaluation budgets, (i) how much of the budget is consumed by stagnation stretches, (ii) how often such stretches end with a real improvement, and (iii) how many of the algorithm's novel evaluations actually connect to improvements. The audit shows that standard GWO is not lacking randomness - its evaluations are the *most* novel of all compared algorithms - but that almost none of this novelty converts into progress, and that its stagnation stretches are both more frequent and much more persistent than those of strong PSO variants.

These findings motivate a deliberately minimal intervention, duty-cycled shell sampling (DCS-GWO). Every generation keeps the GWO equations for most of the pack, but a fixed share θ of the wolves - the least productive ones, ranked by time since their personal record improved - is instead re-deployed to sample a fresh point inside a shell centred on the incumbent α. The shell radius is not a tuned constant: it is estimated online as the median length of the most recent successful displacements, so the sampler inherits its scale from the algorithm's own empirical progress. The roster consumes exactly the same number of true evaluations as the GWO moves it replaces, and the estimates behind the radius are computed from already-evaluated information, so the modification is FE-neutral by construction and adds $O(\theta ND)$ arithmetic per generation.

The contributions of this draft are:

- A reproducible audit of *evaluation productivity* in GWO

Draft produced by the NEW_GWO_2026 project. Method draft; all reported numbers come from the frozen formal batch recorded in the accompanying repository.

and five strong baselines under identical true-FE budgets, including an operational definition of stagnation recovery and of useful novelty that is re-used throughout the paper (Section IV-A).
- DCS-GWO, a one-parameter, FE-neutral modification of GWO whose design follows directly from the audit: duty-cycled, success-scaled shell sampling around the incumbent (Section IV).
- A pre-registered, dual-axis evaluation protocol (stagnation recovery and exploration efficiency, with negative results retained) executed on CEC2017 at $D = 30$ (development) and $D = 50$ with 100,000 true evaluations, 30 runs, and 29 functions (Section V); strong baselines are reused only after being reproduced bit-for-bit on this machine (Section V-B).
- An honest account of what does *not* work: novelty-filtered destinations, self-centred sampling, adaptive quota control, and four stagnation-triggered re-deployment rules all fail pre-registered gates and are reported rather than removed (Sections VI-C and VII).

II. Related Work

A. Grey wolf optimization and its variants

Since [**?**], GWO derivatives have mostly modified the search equations or the social structure: dynamic leaders and reinforcement-learning-selected leadership [**?**], multi-population search with dimension-wise learning and heavy-tailed mutation [**?**], strengthened operators with elite re-selection [**?**], and dynamic grouping strategies. Recent reviews [**?**] find that diversity collapse, local stagnation and the exploration–exploitation balance remain the recurring, unsolved issues, and recent work also warns that mechanism stacking has become the dominant (diminishing-returns) pattern. DCS-GWO deliberately departs from that pattern: the GWO equations are untouched; only the *allocation* of the scarce evaluations is changed.

B. Evaluation-efficient optimization

Sparse-evaluation frameworks evaluate only a subset of the candidates produced per generation; spatial screening and reinforcement-learning scheduling were shown to be effective under heavily constrained budgets [**?**], and behaviour-aware benchmarking has recently been proposed to expose exactly such evaluation dynamics [**?**]. Surrogate-assisted and asynchronous schemes attack the same asymmetry with models or event-driven evaluation. DCS-GWO differs from sparse evaluation in a small but important way: no evaluation is skipped and no candidate generation is added. The same N true evaluations per generation are used, but a fixed fraction of them is moved from leader-following positions to incumbent-anchored sampling. This keeps the method directly comparable under equal-FE protocols, and it isolates the question "which positions deserve the evaluations?" from the question "how many candidates deserve to be evaluated?".

C. Strong baselines

Beyond standard GWO we compare against (i) an author-faithful improved GWO with dimension-wise learning [**?**]; (ii) heterogeneous improved dynamic multi-swarm PSO [**?**]; (iii) heterogeneous comprehensive-learning PSO [**?**]; (iv) dynamic multi-swarm PSO [**?**]; and (v) further established baselines (SPSO-2011 [**?**], CLPSO [**?**], HPSO-TVAC [**?**], competitive swarm optimizer [**?**]). All strong baselines are reused from audited 30-run result tables only after this project reproduced their implementations bit-for-bit under the common protocol (Section V-B).

III. Preliminaries

A. Standard GWO

Given N wolves with positions $x_i \in \mathbb{R}^D$, GWO evaluates the initial population, ranks the three best wolves α, β, δ, and repeats two phases. The exploration parameter shrinks linearly, $a(t) = 2 - 2\,\mathrm{FE}(t)/B$ where $\mathrm{FE}(t)$ is the number of true evaluations consumed and B the budget. For every wolf i (in index order), coefficient vectors are drawn per dimension, $\mathbf{A}_k = 2a\mathbf{r}_1 - a$, $\mathbf{C}_k = 2\mathbf{r}_2$ with $\mathbf{r}_1, \mathbf{r}_2 \sim U(0,1)^D$, and

$$x_i' = \frac{1}{3} \sum_{k \in \{\alpha,\beta,\delta\}} \left(x_k - \mathbf{A}_k \odot |\mathbf{C}_k \odot x_k - x_i|\right), \qquad (1)$$

followed by clipping to the box domain and one true evaluation (leaders are refreshed immediately, following the reference implementation). Each generation therefore consumes exactly N true evaluations; the initialisation consumes N more, and the budget B is enforced exactly in the implementation.

B. Operational audit metrics

All statements in this paper about stagnation and discovery refer to the following operational, algorithm-independent definitions. Let $T(f)$ be the best objective value among the first f true evaluations, and let the stagnation window be $W = \max(50, \lceil 0.01B \rceil)$. A *stagnation episode* is a maximal run of evaluations without a strict improvement of T whose length is at least W; the *recovery probability* is the fraction of stagnation episodes followed by a real improvement within $R = 4W$ further evaluations, and the *long-stagnation share* is the fraction of the budget spent inside such episodes. An evaluation is *novel* if its normalised nearest-neighbour distance to the previously evaluated positions, sampled uniformly with reservoir sampling, exceeds $\tau = 1\%$ of the search-space diagonal; *useful novelty* is the fraction of novel evaluations that are the most recent novel point before a real improvement within W evaluations. These definitions are pre-registered (docs/08 in the accompanying repository) and are applied identically to every algorithm through a common evaluation-interface probe that also audits the true-FE count independently of the algorithm's own counter.

IV. Proposed Method

DCS-GWO modifies exactly one decision variable of GWO: *which positions receive the N true evaluations of each generation, and from which wolves those positions are produced.*

TABLE I
BEHAVIOUR AUDIT (DEVELOPMENT SET, $D = 30$, 30K TRUE FE). STAGNATION WINDOW $W = 300$; NOVELTY THRESHOLD 1% OF THE DIAGONAL.

Algorithm	Recov. prob.	Long-stag. share	Useful nov.	Rank
GWO	0.775	0.862	0.0095	5.6
IGWO [**?**]	0.791	0.851	0.0101	3.8
HIDMS-PSO [**?**]	0.893	0.470	0.0510	2.4
HCLPSO [**?**]	0.966	0.359	0.0572	2.5
DMS-PSO [**?**]	0.855	0.234	0.0919	2.4
SAQ-CSO	0.898	0.734	0.0305	4.4

The GWO equations remain in place for the remaining wolves, the population size, the a-schedule, the leadership rule and the budget accounting are untouched, and no surrogate model, restart or additional evaluation is used.

A. Motivating audit

Table I summarises the audit of five strong baselines and standard GWO on the stratified development set ($D = 30$, 30,000 true evaluations, eight functions, five seeds, identical probe for all algorithms). Two facts stand out. First, GWO fails on the *productivity* of its evaluations: it has the highest pure novelty rate of all compared algorithms (0.89) yet the lowest useful-novelty rate (below 1%), about ten times below DMS-PSO. Its evaluations are spread out, but they do not convert. Second, its stagnation profile is the worst of the set: 86% of the budget is spent inside episodes of at least 1% of the budget and the median recovery needs about 500 evaluations, three times HCLPSO's 145. The two audit axes therefore point to the same prescription: keep the evaluations that refine the incumbent basin, but stop spending a large, fixed share of them on positions that merely confirm it.

B. Duty roster

Every generation, the wolves are ranked by two keys: personal-record staleness (true evaluations since the wolf's own best objective value last improved, decreasing) and personal best value (worse first). A fixed number $\lceil \theta N \rceil$ of wolves from the *lower half* of the personal record ranking is assigned to shell sampling; the upper half always performs the GWO move. The lower-half restriction follows the empirical rule observed in restart-based methods: re-deploying productive wolves destroys ongoing convergence, while the least productive wolves are the cheapest to re-purpose. θ is a single constant (20% in the frozen version) and its value was selected on the development set by pre-registered comparisons against 5%, 10%, 30%, random and adaptive quotas; adaptivity did not pass its gate and was dropped (Section VI-C).

C. Shell sampling with a success-scaled radius

A rover does not follow the leaders. It draws one candidate uniformly inside a per-dimension box of half-width σ centred on the incumbent α,

$$x' = \alpha + \sigma\,(2\mathbf{u} - 1), \qquad \mathbf{u} \sim U(0,1)^D, \tag{2}$$

Algorithm 1 DCS-GWO (one generation, N true evaluations)

```
Rank wolves; choose roster R (|R| = ⌈θN⌉) from the lower half by decreasing staleness.
σ ← success-scale of Eq. (3)
for each wolf i = 1, ..., N do
    if i ∈ R then
        x'_i ← clip(α + σ(2u − 1))   ▷ shell sample, 1 true evaluation
    else
        x'_i ← standard GWO move of Eq. (1)   ▷ 1 true evaluation
    end if
    v ← f(x'_i); refresh α, β, δ; update personal record and S
end for
```

clipped to the domain. The scale is learned online from the algorithm's own progress: let $\mathcal{S}$ be the 25 most recent displacement lengths $\|x'_i - x_i\|$ recorded at true evaluations that strictly improved a wolf's personal record; then

$$\sigma = \max(\mathrm{median}(\mathcal{S}),\, 0.01\,\|u_b - l_b\|) \Big/ \sqrt{D/3}, \tag{3}$$

where the normalisation converts the expected displacement of a D-dimensional uniform box sample back to the recorded displacement scale, and the floor prevents the shell from collapsing to numerical noise. The radius therefore tracks the algorithm's empirical notion of "a step that recently worked", without a hand-tuned decay schedule. The rationale is partially derived from the audit and partially from the earlier observation reproduced here on the development set: evaluations near the incumbent convert to improvements at a rate an order of magnitude higher than far-away evaluations, whereas purely novel points (the maximum-novelty pick tested in the ablation) waste FE at exactly the observed GWO baseline rate.

D. Algorithm and budget accounting

Algorithm 1 lists one generation. Initialisation follows standard GWO. The total number of true evaluations is exactly B (budget) for any N, θ, identical to GWO; the roster decision and the scale estimate use only already-computed quantities. The per-generation overhead is $O(N \log N)$ for the ranking and $O(\theta N D)$ for the sampling, i.e., the same order as one GWO update; on CEC2017 at $D = 50$ the measured wall-clock ratio against GWO is 1.04 on the 29-function average and 1.32 on F_1 (Section VI-E). No distance-to-archive computation is used in the frozen method.

V. EXPERIMENTAL SETUP

A. Protocol

All experiments follow a single equal-FE protocol: CEC2017 functions [**?**], F_1 and F_3–F_{30} (29 functions, F_2 excluded by convention), $B = 100{,}000$ true objective evaluations including initialisation, 30 independent runs (seeds 3001–3030), $D = 50$ for the formal tier. Every objective call of every algorithm passes through a common probe that

counts evaluations independently; a run is discarded if the probe's count differs from the algorithm's own counter or from B. The initial population of each (function, seed) pair is drawn once from a fixed protocol seed and shared as a prefix by all algorithms, so paired comparisons use identical starting points. All runs execute on four local MATLAB workers (Parallel Computing Toolbox, `parfor` over independent algorithm×function×seed jobs); each finished run is written to disk immediately, and no serial fallback is used.

B. Audited baselines

Strong baselines are reused from audited 30-run tables only because this project reproduces them *bit-for-bit* under the same protocol: for each baseline we reran sample (function, seed) pairs and required agreement to machine precision with the archived results (maximum relative difference $< 3 \times 10^{-15}$ over the checked pairs). The audited entries are: IGWO (author-faithful dimension-wise learning) [**?**], HIDMS-PSO [**?**], HCLPSO [**?**], DMS-PSO [**?**], SAQ-CSO (sparse-evaluation CSO with learned quotas), SPSO-2011 [**?**], CLPSO [**?**], HPSO-TVAC [**?**], and the competitive swarm optimizer [**?**]. Baseline hyper-parameters and populations are the authors'/archives' originals and are not tuned here.

C. Statistics

Pairing unit is (function, seed); comparisons on errors use $\log_{10}$-transformed values and two-sided Wilcoxon signed-rank tests [**?**] per function with Holm correction [**?**] across functions, plus average ranks in the Friedman sense and paired median error ratios; weak comparisons are additionally reported as function-level wins and losses. No per-function result is removed. Development decisions are made on a separate $D = 30$, 30,000-evaluation, eight-function development set with pre-registered gates; the formal $D = 50$ tier is executed once after the method is frozen and is never used for tuning.

VI. Results

A. Formal comparison

Table II reports the formal tier: CEC2017, $D = 50$, 100,000 true evaluations, 29 functions, 30 runs, 15 algorithms. Standard GWO ranks 13th of 15 (mean rank 11.45); DCS-GWO climbs to 7th (8.55), behind the three heterogeneous/quasi-Newton PSO variants and behind IGWO, SAQ-CSO and SPSO-2011, and ahead of HPSO-TVAC, CLPSO and original CSO. Against GWO, DCS-GWO wins 21 of 29 functions with a median paired error ratio of 0.792; after per-function Holm correction 11 functions are significantly better and none significantly worse (pooled Wilcoxon $p \approx 4 \times 10^{-40}$). Table III shows where the gain comes from: the mechanism is very strong on the two unimodal functions (median ratio 0.10) and on the hybrid functions (0.61), moderate on composition functions (0.90) and essentially neutral on the simple multimodal group (1.02), where GWO's many local optima are not the bottleneck that shell sampling addresses.

TABLE II
Formal comparison, CEC2017 $D = 50$, 100k true FE, 29 functions, 30 runs; baseline rows are audited and reproduced bit-for-bit. "Sig." is the number of functions with Holm-adjusted paired Wilcoxon $p < 0.05$.

Algorithm	Mean rank	W/T/L vs GWO	Med. ratio	Sig. (+/−)	Time
HIDMS-PSO	2.52	29/0/0	0.342	27/0	3
HCLPSO	3.17	29/0/0	0.397	29/0	2
DMS-PSO	3.41	28/0/1	0.389	28/1	1
IGWO	5.55	27/0/2	0.465	25/0	2
SAQ-CSO	6.10	29/0/0	0.578	21/0	9
SPSO-2011	6.48	26/0/3	0.552	21/1	2
ANGWO-final	8.55	21/0/8	0.792	11/0	3
ANGWO-ES2	8.83	22/0/7	0.807	12/0	3
HPSO-TVAC	8.90	17/0/12	0.799	13/10	1
ANGWO-adapt	9.00	23/0/6	0.825	8/0	3
CLPSO	9.55	21/0/8	0.789	16/3	2
ANGWO-novdest	9.93	23/0/6	0.830	9/0	4
GWO	11.45	–	1.000	–	3
ANGWO-self	11.59	13/0/16	1.013	0/0	3
Original-CSO	14.97	0/0/29	3.336	0/29	2

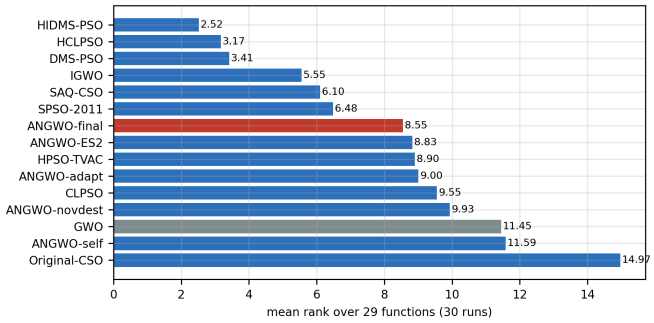


Fig. 1. Mean ranks over 29 functions, CEC2017 $D = 50$, 100,000 true evaluations, 30 runs. DCS-GWO is shown in red, standard GWO in grey.

B. Dimension scalability ($D = 100$)

A companion tier reruns GWO and DCS-GWO at $D = 100$ (100k true FE, 29 functions, 10 seeds) and pairs the result with audited IGWO and RASE-GWO rows on their overlapping 12-function subset. The mechanism keeps its sign but loses strength: 18/11 function wins and a median error ratio of 0.878 against GWO on all 29 functions (no function is significant at 10 seeds), and 1.37/1.44 median ratios against IGWO and the sparse-evaluation RASE-GWO on the 12-function subset (5/12 and 1/12 wins). The wall-clock ratio stays 1.01 because evaluation cost dominates further. We read this as a partial transfer of the mechanism to higher dimension and report it as a limitation rather than a scalability claim.

C. Ablation

Table IV reports the formal ablation. Replacing the incumbent (*alpha*) centre by the wolves' own positions destroys the effect completely (median ratio 1.013 versus GWO, 4/29 wins against the frozen method), which identifies shell placement as the essential ingredient, not exploration per se. Replacing the random destination by the maximum-novelty pool weakens the method (0.830 vs 0.792, $p = 0.003$), and an adaptive quota also weakens it slightly (0.825, $p = 0.037$); both alternatives

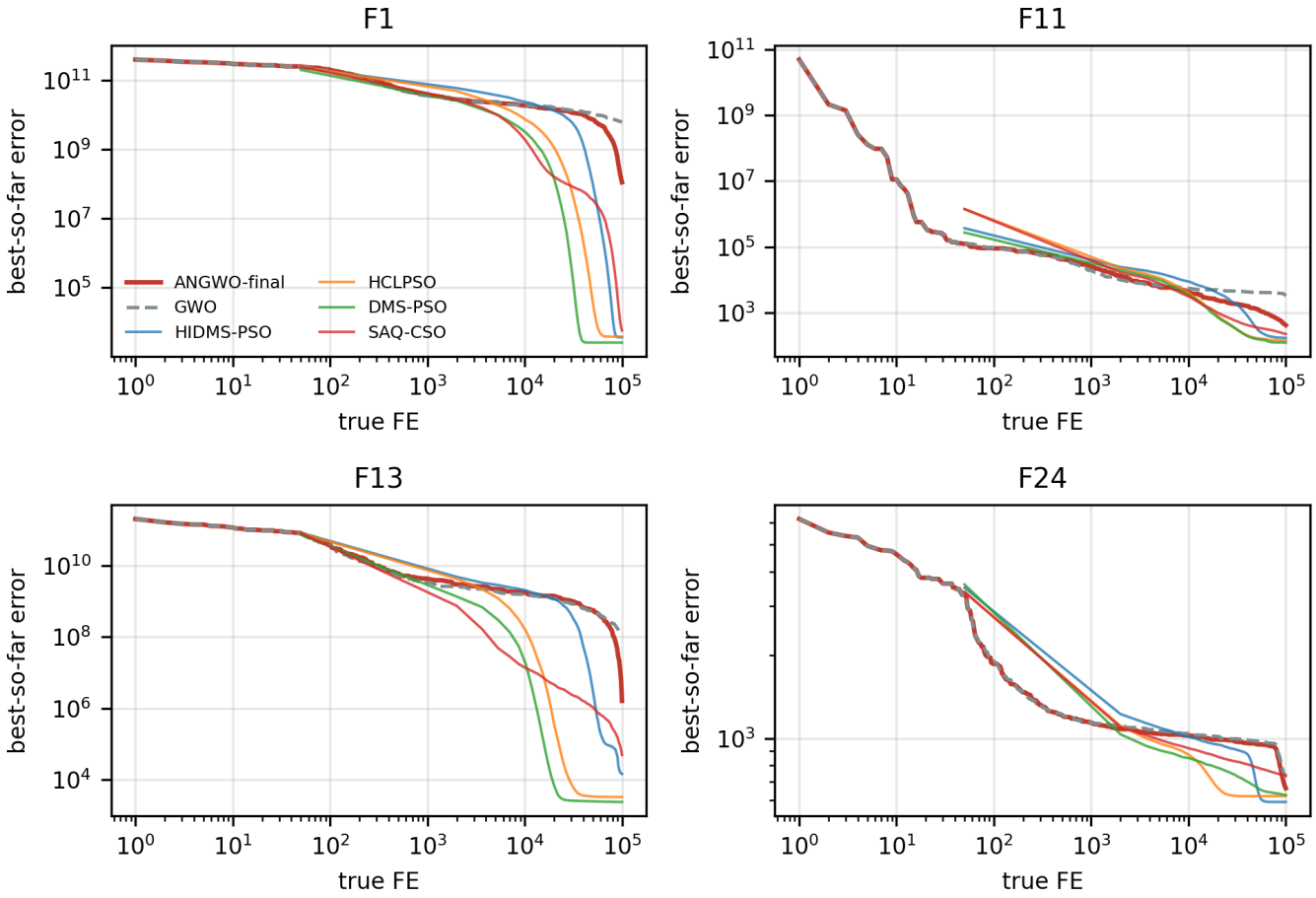


Fig. 2. Median best-so-far error versus true evaluations on four representative functions (F1, F11, F13, F24). Solid red: DCS-GWO; dashed grey: GWO; thin lines: audited strong baselines; all curves share the same true-FE budget.

TABLE III
MEDIAN PAIRED ERROR RATIO OF DCS-GWO OVER GWO BY CEC2017 FUNCTION FAMILY (29 FUNCTIONS).

Family	n	Median ratio
Unimodal (F1, F3)	2	0.102
Simple multimodal (F4–F10)	7	1.024
Hybrid (F11–F20)	10	0.614
Composition (F21–F30)	10	0.896

were rejected by their pre-registered gates and are retained here as negative results. Adding the stagnation-triggered elite-memory re-deployment module changes nothing measurable (0.807, $p = 0.48$), matching its failure on the recovery gate.

D. Behaviour metrics

Table V reports the operational metrics on the formal tier. The picture is consistent with the development screens but more nuanced. The fraction of novel evaluations that connect to a real improvement within the stagnation window rises from 0.0035 (GWO) to 0.0058 (DCS-GWO), a paired median increase of 64% ($p \approx 5 \times 10^{-94}$), and the late-budget novelty rate rises by 15%. The share of the budget spent in long stagnation stretches falls slightly (0.844 to 0.831), but the recovery probability moves only from 0.757 to 0.768, far below the pre-registered gate of $+0.05$. The number of global-best improvement events per evaluation is not increased by the mechanism (median ratio 0.99 overall; the last-quarter ratio is 0.94 on the median but improves strongly on the mean), so the end-point error gains should be read as *better placement of the same number of discoveries* rather than as more discoveries per se.

E. Runtime

Table VI lists wall-clock times measured under identical instrumentation (four workers, same machine). The frozen method costs $1.04\times$ GWO on the 29-function average, and $1.32\times$ GWO on F1 alone; SAQ-CSO costs $2.5\times$ GWO. All runs are exactly 100,000 true evaluations; the method generates 0.2 extra candidate positions per evaluation (the rovers' shell samples that replace GWO moves) and no distance computations.

TABLE IV
FORMAL ABLATION (SAME PROTOCOL AS TABLE II); "VS FINAL" IS THE PAIRED COMPARISON AGAINST DCS-GWO.

Variant	Mean rank	W/T/L vs GWO	Med. ratio vs GWO	vs final (ratio; W/T/L)	p_{final}
GWO	11.45	–	1.000	–	–
ANGWO-final	8.55	21/0/8	0.792	1.000; 0/29/0	nan
ANGWO-self	11.59	13/0/16	1.013	1.285; 4/0/25	5.85e-44
ANGWO-novdest	9.93	23/0/6	0.830	1.030; 9/0/20	2.99e-03
ANGWO-adapt	9.00	23/0/6	0.825	1.045; 9/0/20	3.72e-02
ANGWO-ES2	8.83	22/0/7	0.807	0.998; 14/0/15	4.76e-01

TABLE V
BEHAVIOUR METRICS ON THE FORMAL TIER (DCS-GWO VARIANTS AND GWO; AUDITED BASELINES DO NOT CARRY THE PROBE RECORD).

Algorithm	Recov. prob.	Long-stag. share	Impr. rate Q4	Useful nov.	Nov. rate Q4
ANGWO-final	0.768	0.831	0.0045	0.0058	0.690
ANGWO-ES2	0.766	0.832	0.0045	0.0057	0.691
ANGWO-adapt	0.767	0.828	0.0050	0.0061	0.683
ANGWO-novdest	0.765	0.830	0.0046	0.0065	0.704
GWO	0.757	0.844	0.0048	0.0035	0.588
ANGWO-self	0.761	0.847	0.0044	0.0059	0.681

TABLE VI
WALL-CLOCK PER RUN (MEAN OVER 870 RUNS PER ALGORITHM) AND CANDIDATE-GENERATION COST. "CANDIDATES/FE" COUNTS GENERATED POSITIONS THAT RECEIVE A TRUE EVALUATION OR REPLACE ONE.

Algorithm	Time (s)	Ratio vs GWO	Candidates / FE
HIDMS-PSO	3.32	0.91	n/a
HCLPSO	2.13	0.59	n/a
DMS-PSO	1.89	0.52	n/a
IGWO	2.32	0.64	n/a
SAQ-CSO	9.08	2.50	n/a
SPSO-2011	2.56	0.71	n/a
ANGWO-final	3.76	1.04	0.200
ANGWO-ES2	3.86	1.06	0.214
HPSO-TVAC	1.80	0.50	n/a
ANGWO-adapt	3.74	1.03	0.070
CLPSO	2.41	0.66	n/a
ANGWO-novdest	4.94	1.36	1.599
GWO	3.63	1.00	n/a
ANGWO-self	3.76	1.04	0.200
Original-CSO	2.24	0.62	n/a

VII. DISCUSSION

A. What the evidence supports

The formal tier supports three claims. (1) The frozen mechanism improves standard GWO robustly: 21/29 function wins, median error ratio 0.792, 11 Holm-significant functions and none significantly worse, at 1.04× GWO's wall-clock and identical true-FE budgets. (2) The gain is *localised*, as the ablation and the family breakdown show: moving the sampling centre from the incumbent to the wolves themselves removes essentially all of the benefit, and the benefit concentrates on unimodal and hybrid landscapes. (3) Per-evaluation *productivity* improves: the fraction of novel evaluations that connect to real improvements is 64% higher in paired medians.

B. The stagnation-recovery axis is not supported

None of six pre-registered stagnation-triggered redeployment variants (far novelty, uniform random, periodic, elite-memory shells, alpha-shell $\times 1/\times 3$, development tier) improved the operational recovery probability by the required five percentage points, and the best of them moved it by $+0.011$ on the formal tier. A periodic variant that spent roughly a third of its budget on re-deployments performed no better, indicating that where a few wolves are teleported matters more than how often, and that under GWO's attraction structure the probability of beating the incumbent within four stagnation windows is governed mainly by how evaluations are placed *near the incumbent* - exactly the channel the frozen method uses. We therefore claim the exploration-efficiency axis and report the recovery axis as unsupported.

C. Limitations

(i) *Gap to the strongest baselines.* DCS-GWO remains behind HIDMS-PSO, HCLPSO, DMS-PSO (median ratios 1.9–2.0), and behind IGWO, SAQ-CSO and SPSO-2011 (1.3–1.4); the paper claims a mechanism and honest evidence, not state-of-the-art dominance. (ii) *Family dependence.* Simple multimodal functions are neutral; a reader should not expect gains there. (iii) *Scale definition.* The shell radius is defined through recorded successful step lengths; other reasonable definitions were not exhausted. (iv) *Benchmark scope.* Only CEC2017 is used; the D=100 tier (accompanying repository) extends the same protocol to one higher dimension but the study still lacks a real engineering problem. (v) *CPU.* The method is FE-neutral but not free: about 4% more wall-clock on average.